05 OCT 2026

25ECPCC305IAE-II

Seat No. : .....

Student Name : .....

St. John College of Engineering and Management, Palghar,

Maharashtra

Autonomous Institute

Internal Assessment Examinations, October 2026

B.Tech. Electronics & Computer Science - SEMESTER III

25ECPCC305 - Data Structure & Algorithms using Python

Time : 1 Hour **Maximum Marks : 20**

Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

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Q1	Attempt any 5 out of 6 Questions	Max. marks(5)	Course Outcome	RBT Level
	a)What type of graph commonly represents road networks?	1	CO4	L1
	b)What is the connection between two vertices called?	1	CO4	L1
	c)What is an occurrence of two keys mapping to the same location called?	1	CO5	L1
	d)Which sorting algorithm uses the divide-and-conquer technique by dividing and then merging?	1	CO5	L1
	e)What property states that an optimal solution contains optimal solutions to subproblems?	1	CO6	L2
	f)If an item has profit 60 and weight 10, what is its profit/weight ratio?	1	CO6	L2
Q2	a)Construct an expression tree for the following expression: (A+B)*(C-D). Then find: Inorder traversal, Preorder traversal, Postorder traversal	5	CO4	L3
	OR			
	b)Construct a BST for the following sequence: 45, 25, 65, 15, 35, 55, 75, 10, 20, 30, 40. Also Delete the nodes 25 and 65	5	CO4	L3
Q3	a)Consider a hash table of size 10 and: h(k)=k % 10 Insert: 23, 43, 13, 33; using Quadratic Probing. Show the position calculated for each collision.	5	CO5	L4
	OR			
	b)Write Python functions for iterative, Binary Search. Analyse: 1. Number of comparisons	5	CO5	L4

	2. Execution approach 3. Additional memory			
Q4	a)Define Backtracking. Explain its basic principle with a suitable example.	5	CO6	L2
	OR			
	b)Given: N = 39 Use bit manipulation to: 1. Check whether bit 2 is set. 2. Set bit 1. 3. Clear bit 0. 4. Toggle bit 3. Show the result after each operation.	5	CO6	L2

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